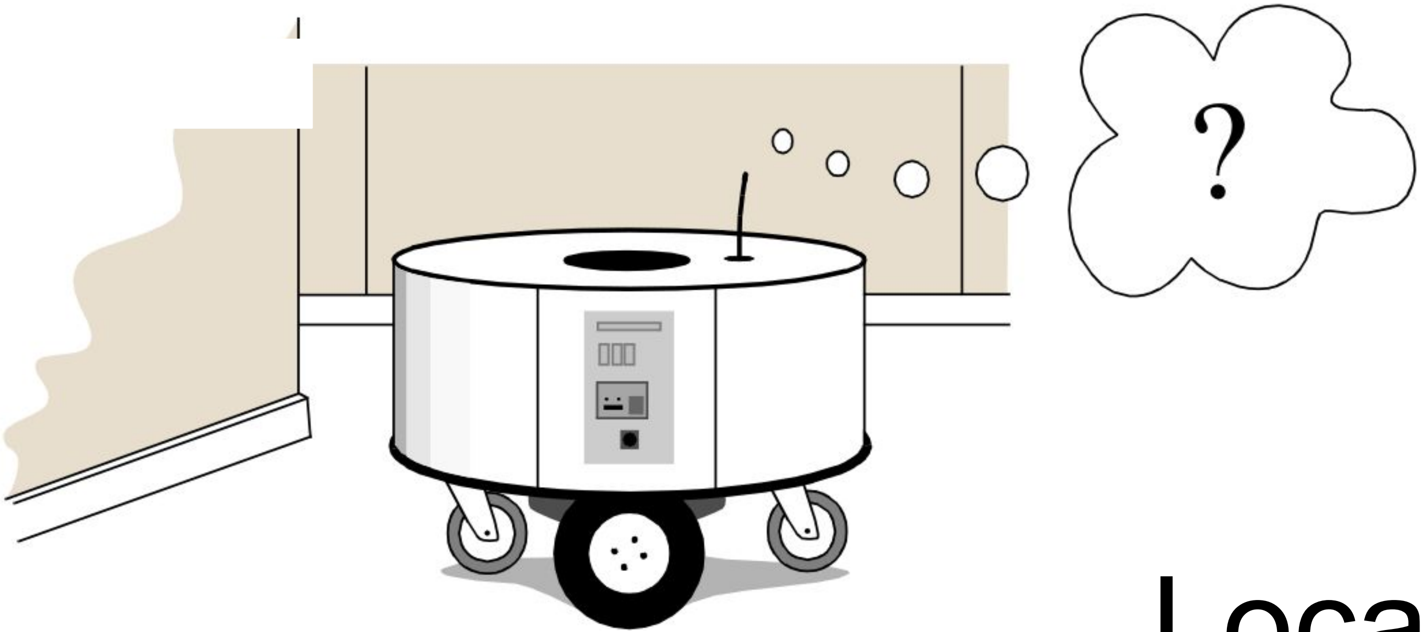
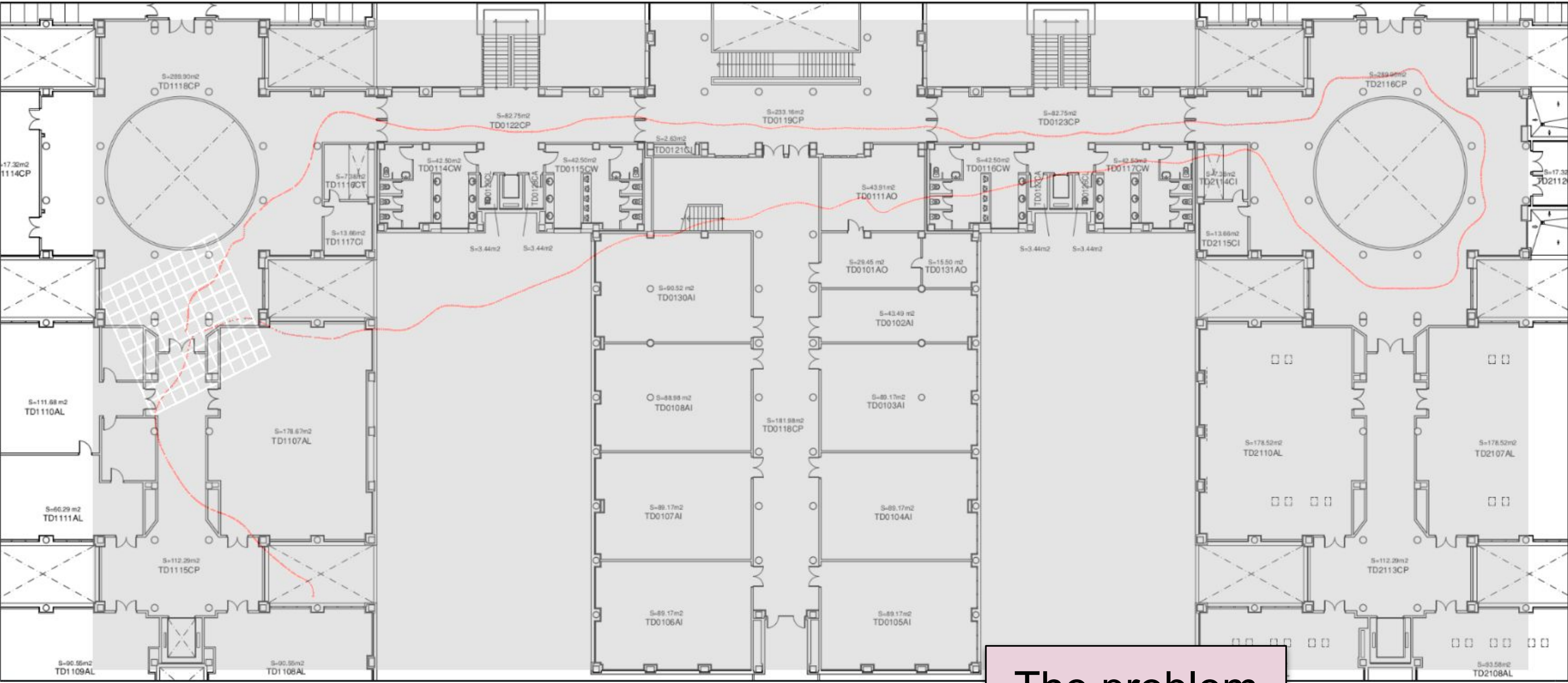
Mobile Robotics

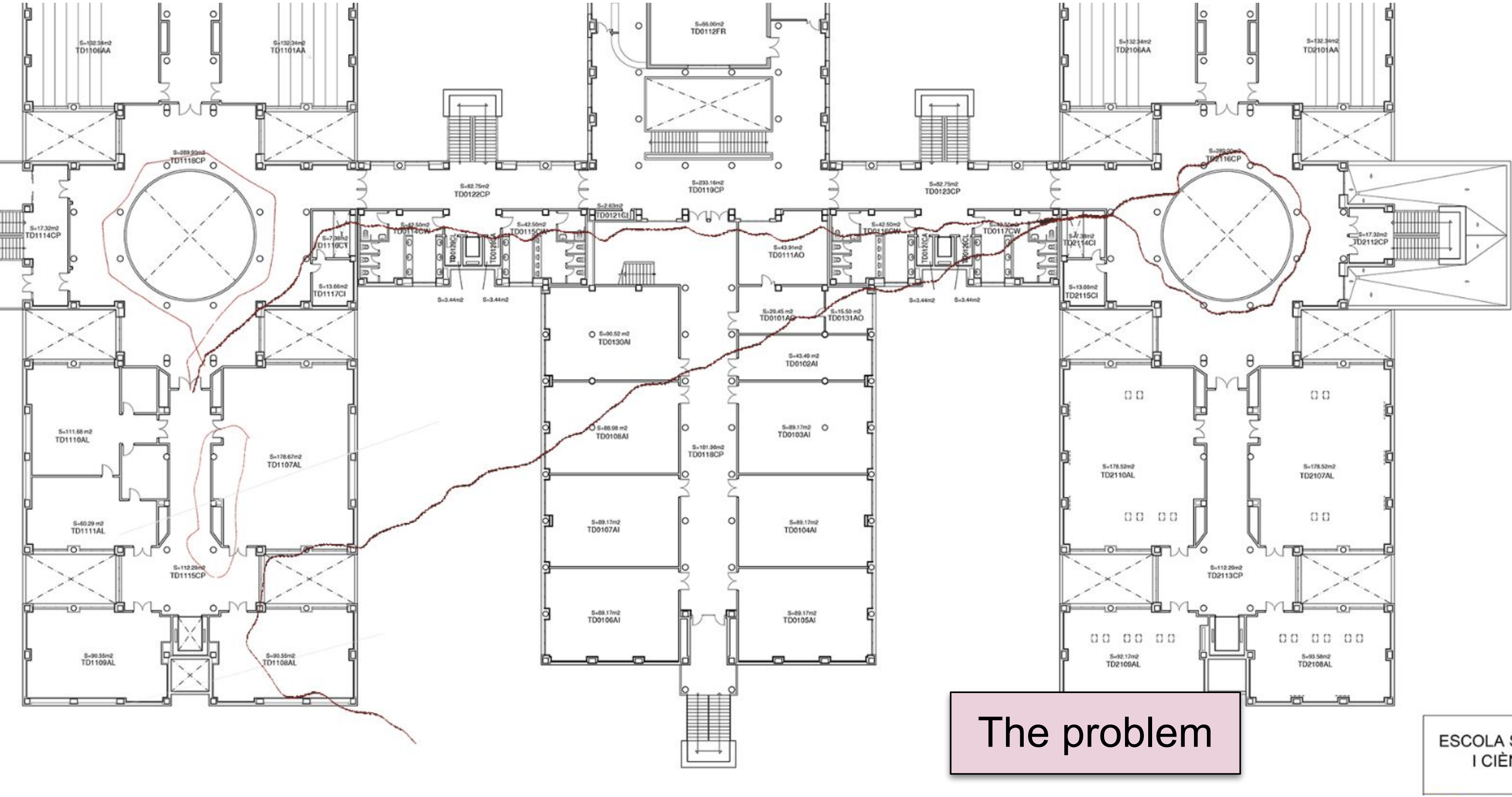
Bachelor's Degree in
Robotic Intelligence



Localization


TD1118CP
TD0122CP
TD0119CP
TD0123CP
TD2116CP
TD0114CW
TD0115CW
TD0116CW
TD0117CW
TD0111AO
TD0101AO
TD0131AO
TD0130AI
TD0102AI
TD0108AI
TD0103AI
TD0118CP
TD0107AI
TD0104AI
TD0106AI
TD0105AI
TD1110AL
TD1107AL
TD1111AL
TD1115CP
TD1109AL
TD1108AL
TD2110AL
TD2107AL
TD2113CP
TD2108AL
TD1117CI
TD2115CI

The problem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TD1101AA
TD0112FR
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TD1118CP
TD0122CP
TD0119CP
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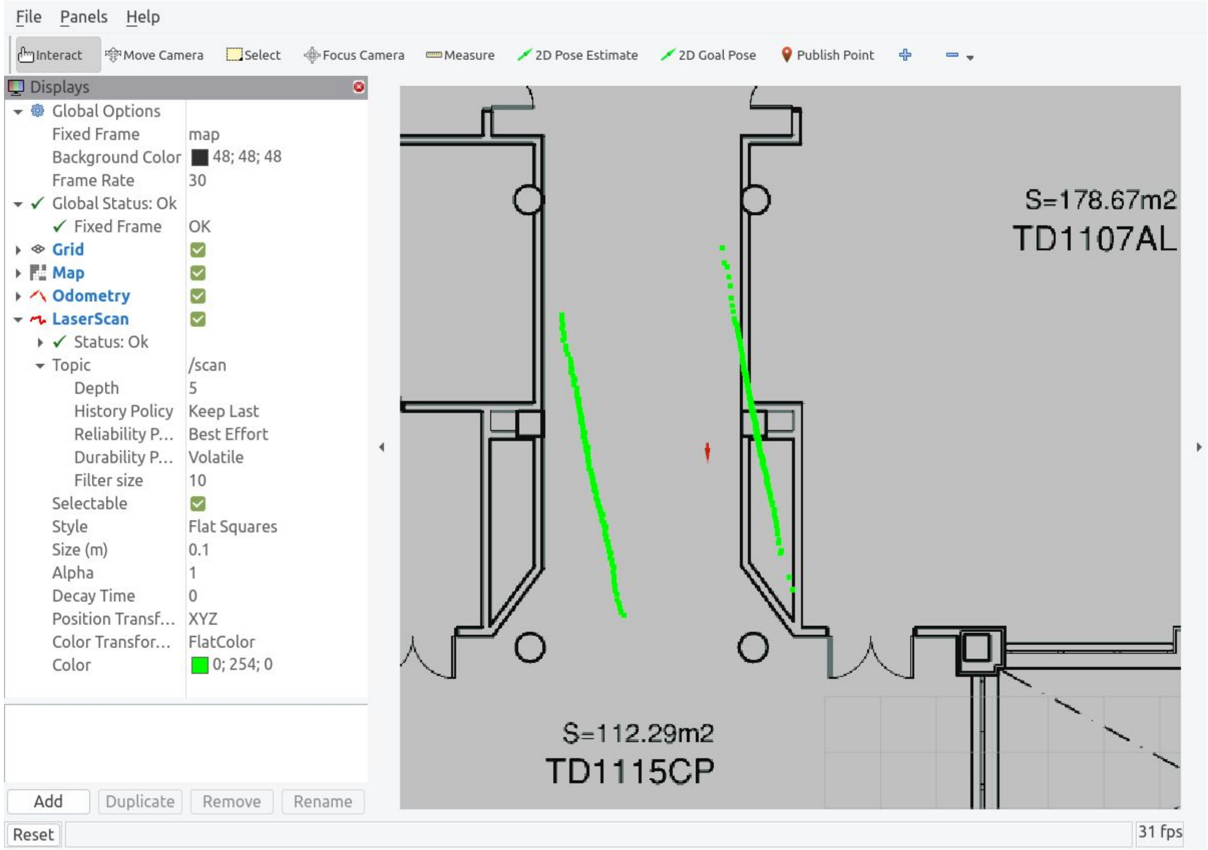
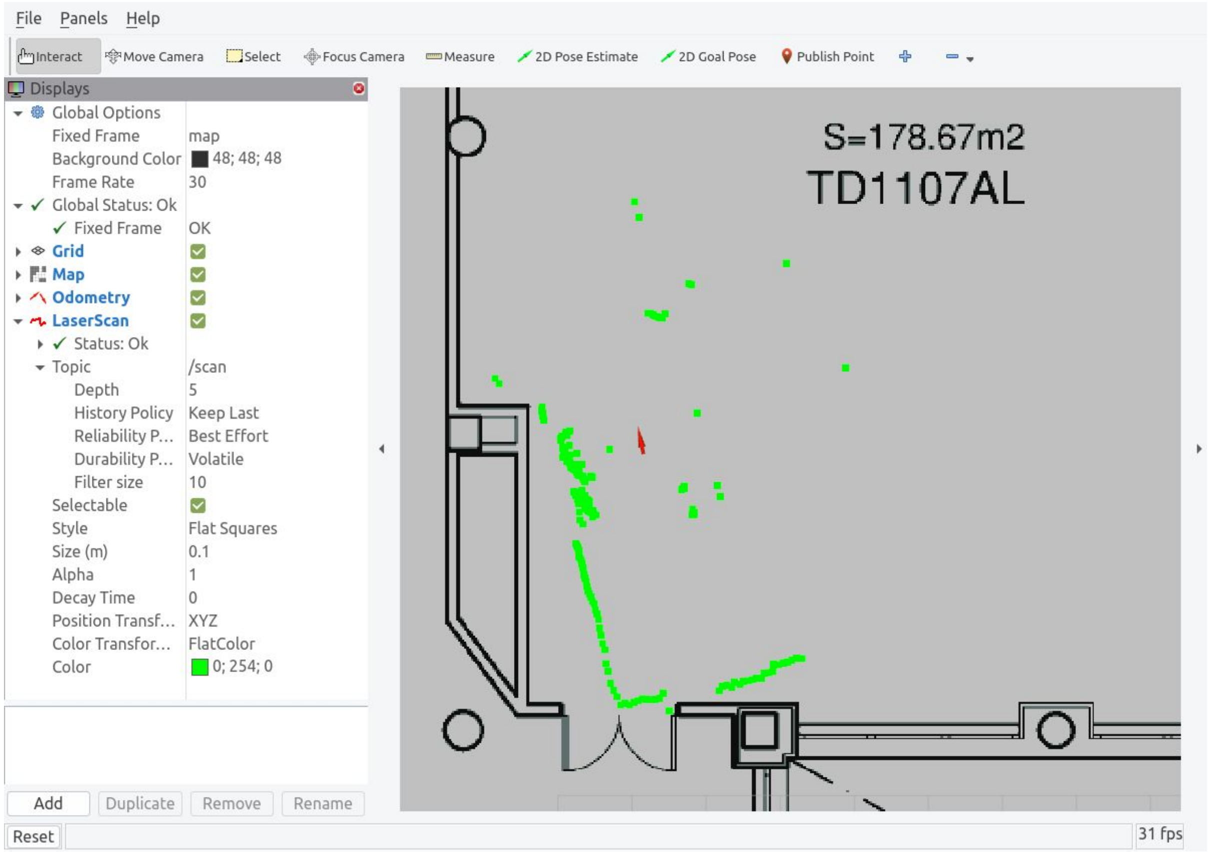
TD0121CI
TD0114CW
TD0115CW
TD0116CW
TD0117CW
TD1116CI
TD2114CI
TD2112CP
TD0111AO
TD1117CI
TD2115CI
TD0101AO
TD0131AO
TD0130AI
TD0102AI
TD0108AI
TD0103AI
TD1110AL
TD1107AL
TD0118CP
TD2110AL
TD2107AL
TD1111AL
TD0107AI
TD0104AI
TD1115CP
TD2113CP
TD0106AI
TD0105AI
TD1109AL
TD1108AL
TD2109AL
TD2108AL

The problem



File
Panels
Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
Displays
Global Options
Fixed Frame
map
Background Color
48; 48; 48
Frame Rate
30
Global Status: Ok
Fixed Frame
OK
Grid
Map
Odometry
LaserScan
Status: Ok
Topic
/scan
Depth
5
History Policy
Keep Last
Reliability P...
Best Effort
Durability P...
Volatile
Filter size
10
Selectable
Style
Flat Squares
Size (m)
0.1
Alpha
1
Decay Time
0
Position Transf...
XYZ
Color Transfor...
FlatColor
Color
0; 254; 0
Add
Duplicate
Remove
Rename
Reset
31 fps
S=178.67m2
TD1107AL
S=112.29m2
TD1115CP

The problem


File
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Move Camera
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Focus Camera
Measure
2D Pose Estimate
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Publish Point
Displays
Global Options
Fixed Frame map
Background Color 48; 48; 48
Frame Rate 30
Global Status: Ok
Fixed Frame OK
Grid
Map
Odometry
LaserScan
Status: Ok
Topic /scan
Depth 5
History Policy Keep Last
Reliability P... Best Effort
Durability P... Volatile
Filter size 10
Selectable
Style Flat Squares
Size (m) 0.1
Alpha 1
Decay Time 0
Position Transf... XYZ
Color Transfor... FlatColor
Color 0; 254; 0
Add
Duplicate
Remove
Rename
Reset
31 fps
S=178.67m2
TD1107AL

The problem

Specification -

Coordinate Frames

base_link

The coordinate frame called `base_link`
position or orientation; for every hardware
103 [1] specifies a preferred orientation fo

odom

The coordinate frame called `odom` is a wo
drift makes the `odom` frame useless as a
meaning that the pose of a mobile platfor

In a typical setup the `odom` frame is comp

drift

noun [countable]

/drɪft/

a slow change, often toward sth bad

decaimiento [masculine]

- *the drift toward war*
  el decaimiento hacia la guerra
- *the Democrats' drift toward the center*
  el decaimiento de los demócratas hacia el centro

a slow movement

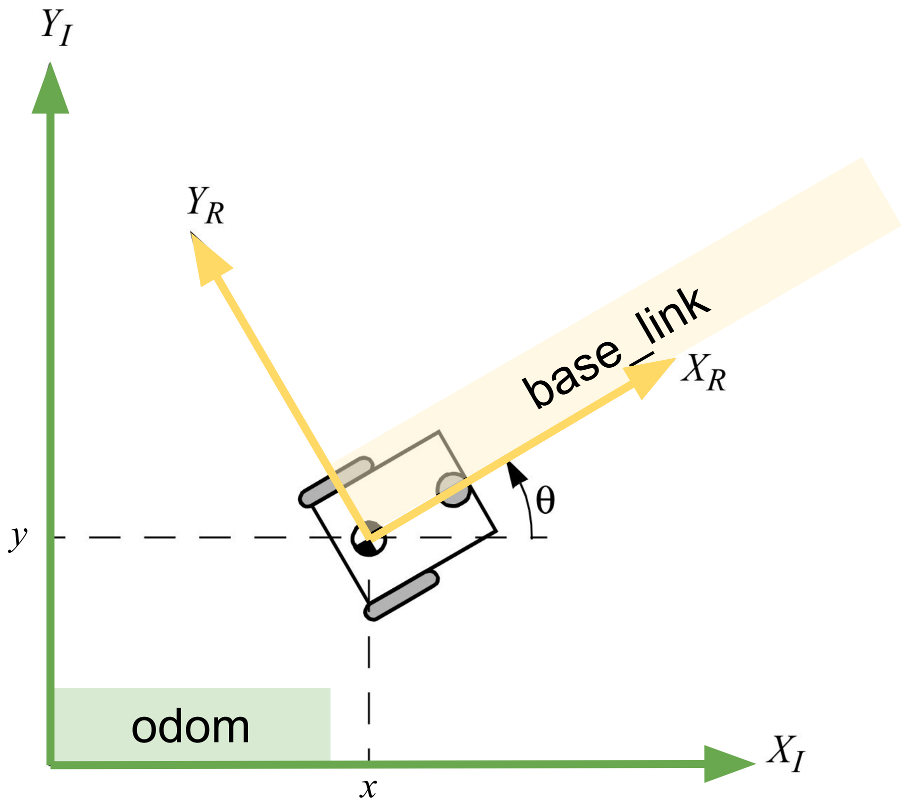
tendencia [feminine]

movimiento [masculine]

- *the drift of the continents*
  el movimento de los continentes

The `odom` frame is useful as an accurate, short-term local reference, but drift makes it a poor frame for long-term reference.

ROS frames of reference

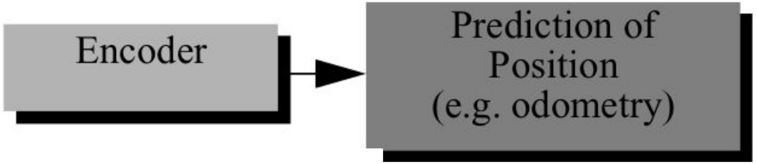


The coordinate frame called **odom** is a world-fixed frame. The pose of a mobile platform in the odom frame can drift over time, without any bounds. This drift makes the odom frame **useless as a long-term global reference**. However, the pose of a robot in the odom frame is guaranteed to be continuous, meaning that the pose of a mobile platform in the odom frame always evolves in a smooth way, without discrete jumps.

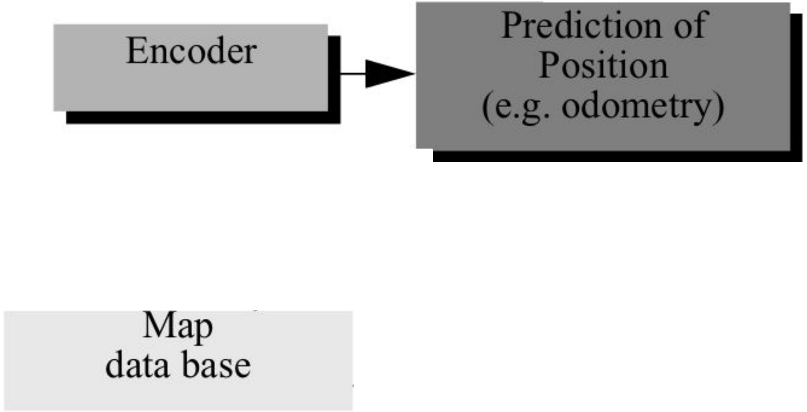
In a typical setup the odom frame is computed based on an **odometry source**, such as **wheel** odometry, **visual** odometry or an **inertial measurement unit**.

The odom frame is useful as an **accurate, short-term** local reference, but drift makes it a **poor frame for long-term** reference.

The solution

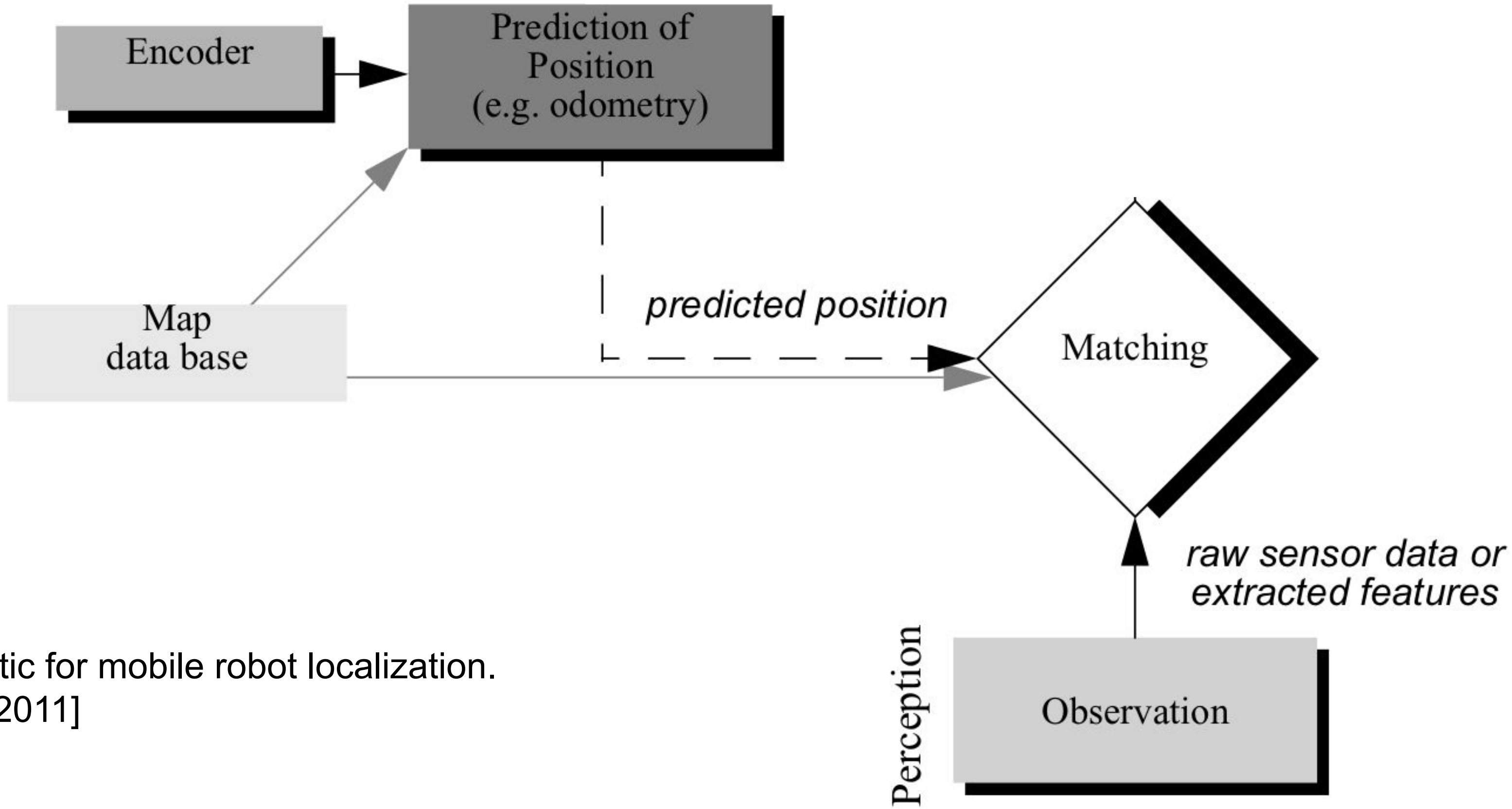


The solution



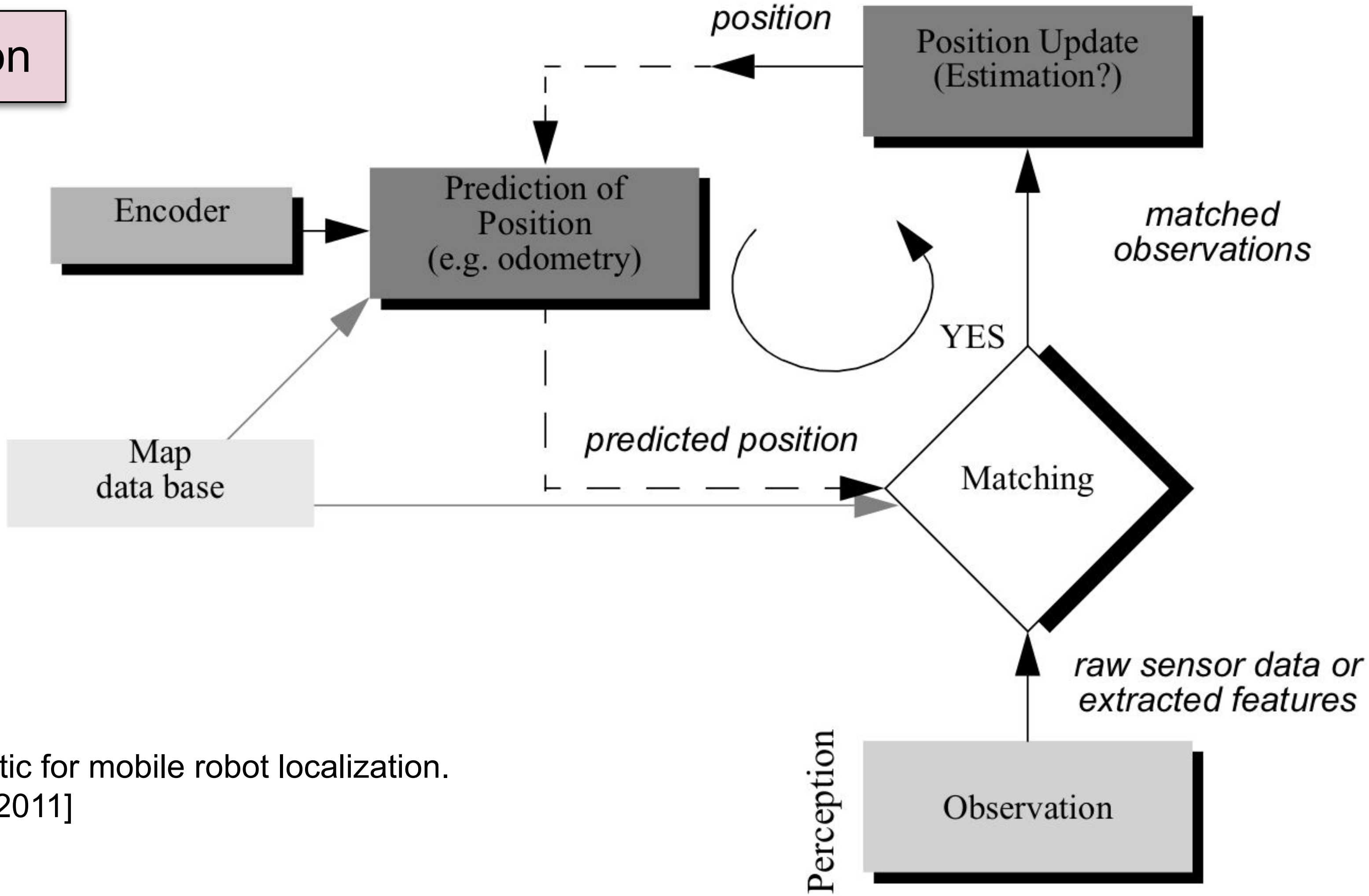
General schematic for mobile robot localization.
[Siegwart et al., 2011]

The solution



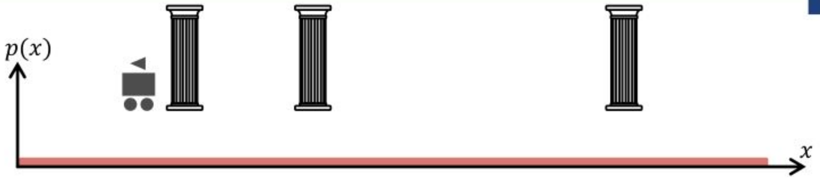
General schematic for mobile robot localization.
[Siegwart et al., 2011]

The solution



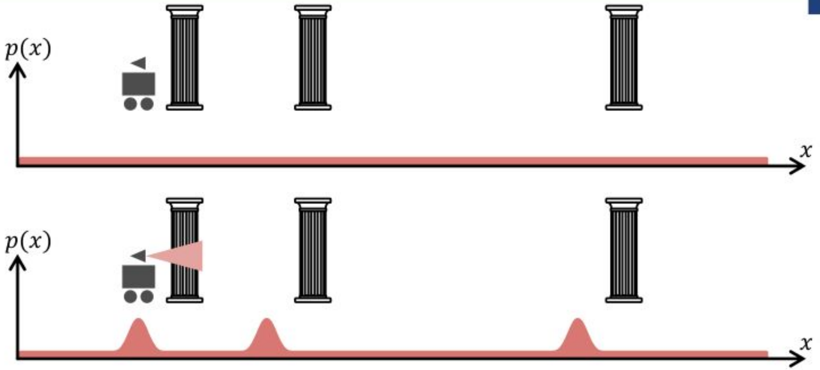
General schematic for mobile robot localization.
[Siegwart et al., 2011]

ACT - SEE Cycle for Localization



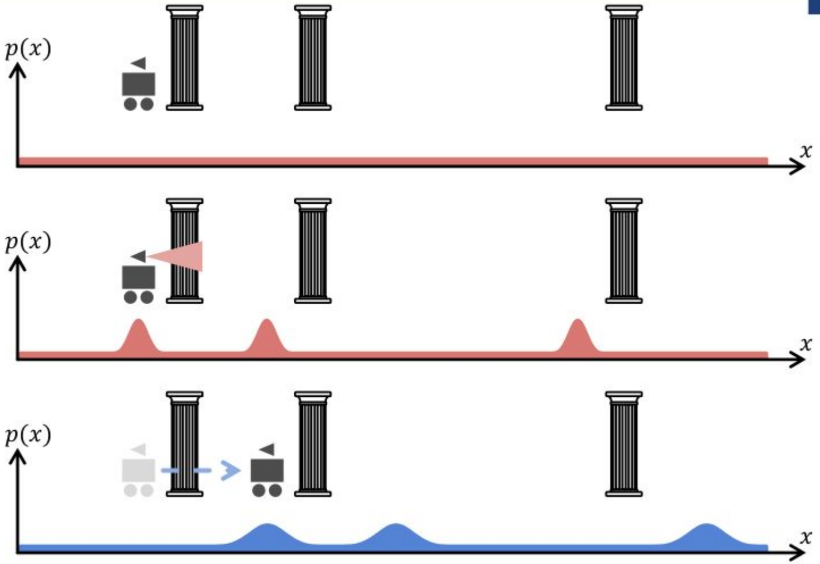
ACT - SEE Cycle for Localization

- SEE: The robot queries its sensors
  → finds itself next to a pillar



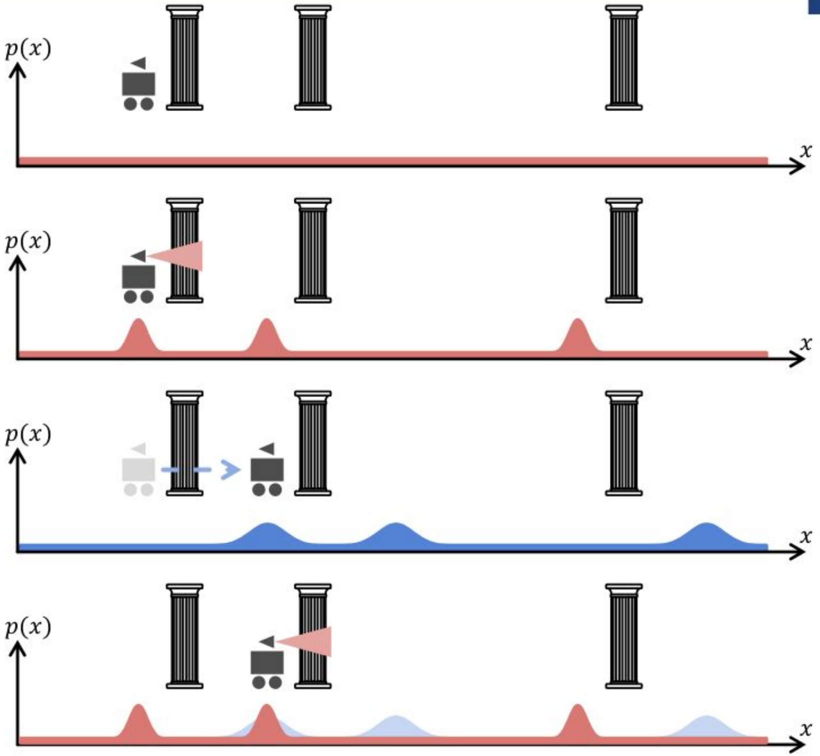
ACT - SEE Cycle for Localization

- SEE: The robot queries its sensors
  → finds itself next to a pillar
- ACT: Robot moves one meter forward
  - motion estimated by wheel encoders
  - accumulation of uncertainty



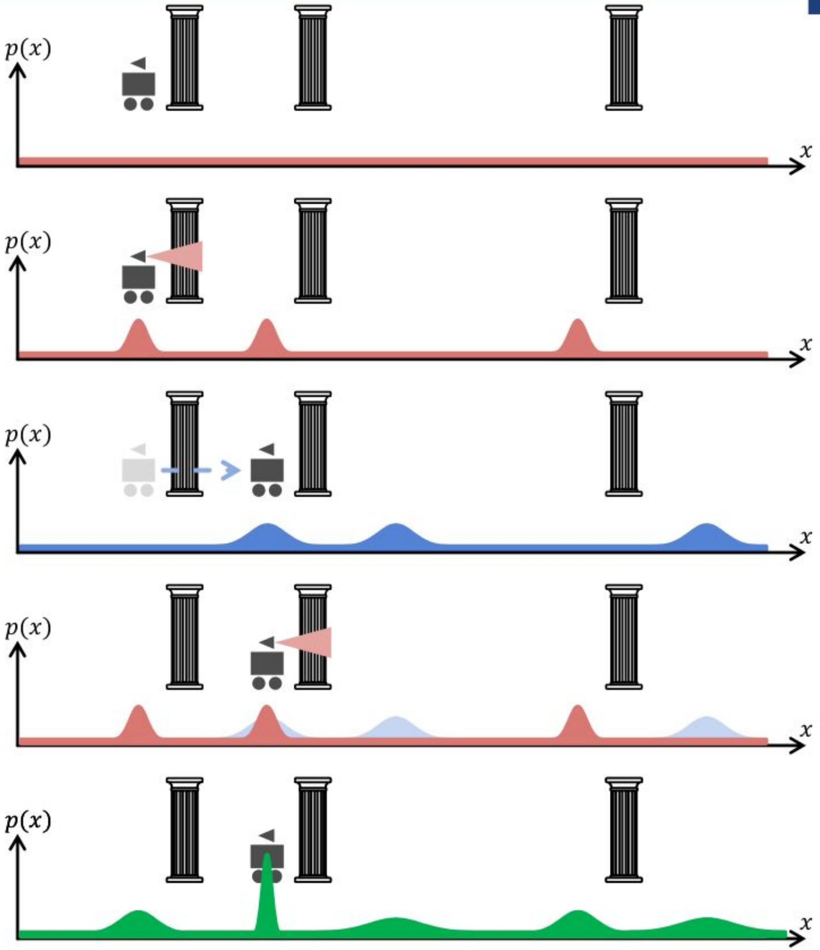
ACT - SEE Cycle for Localization

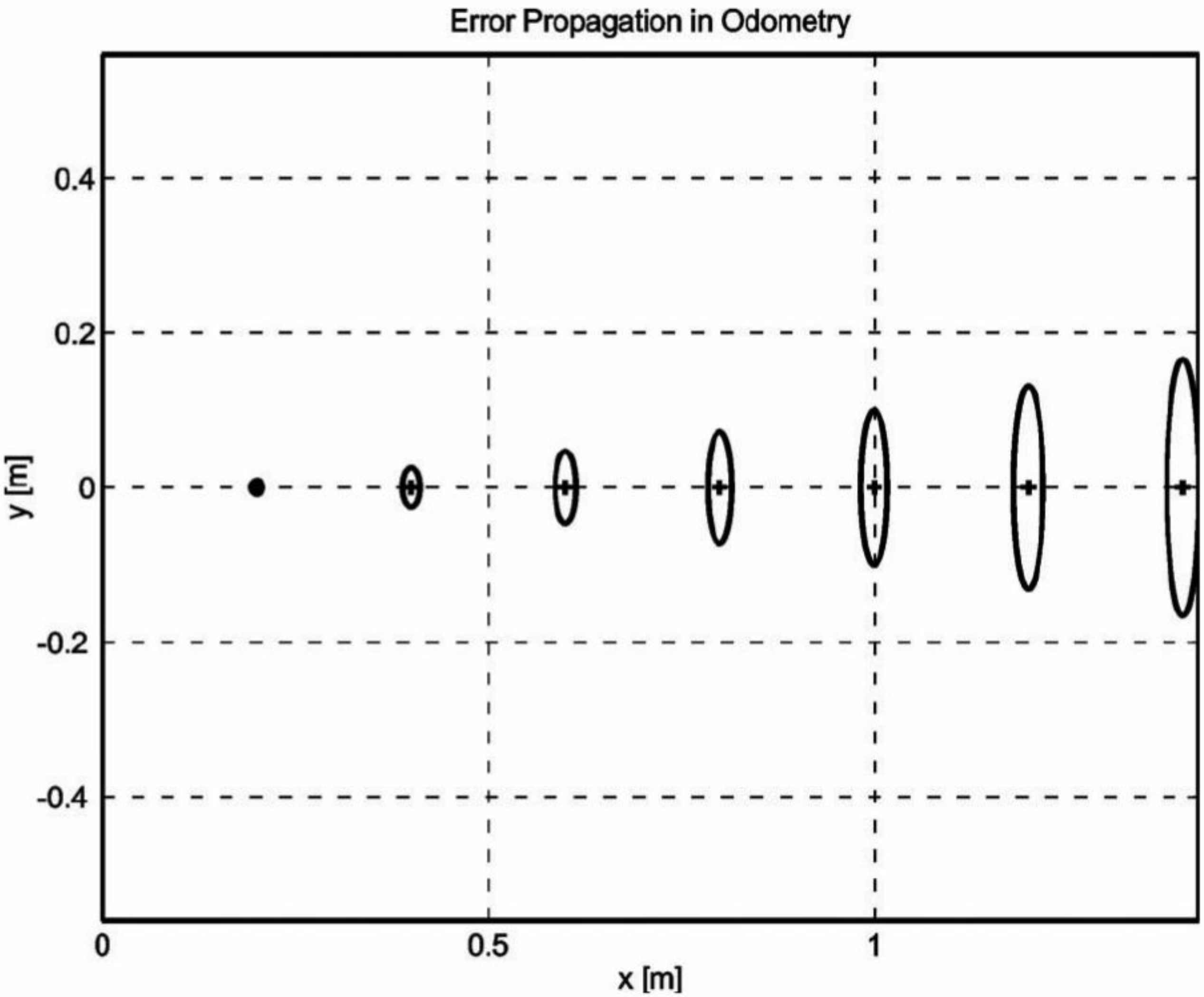
- SEE: The robot queries its sensors → finds itself next to a pillar
- ACT: Robot moves one meter forward
  - motion estimated by wheel encoders
  - accumulation of uncertainty
- SEE: The robot queries its sensors again → finds itself next to a pillar



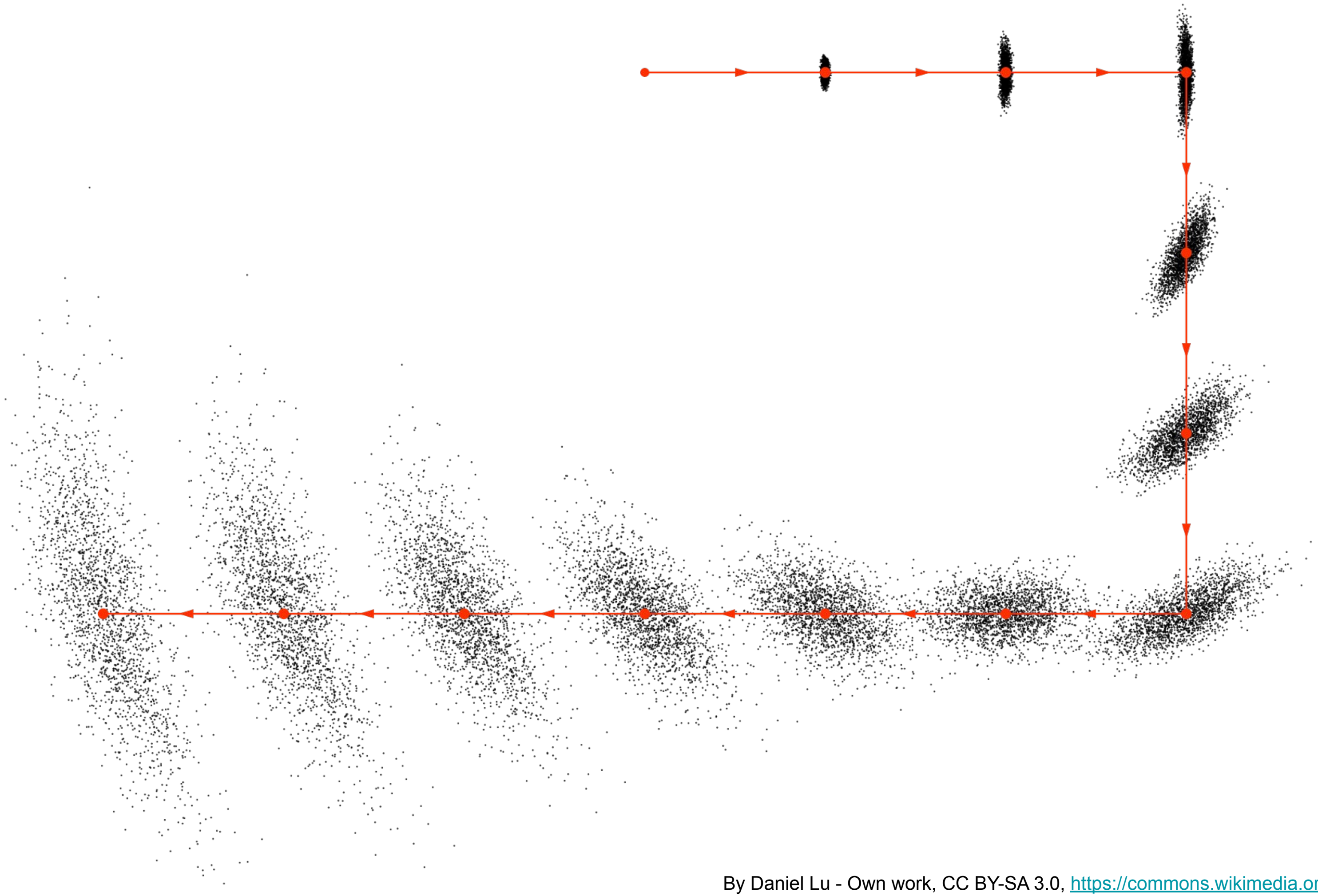
ACT - SEE Cycle for Localization

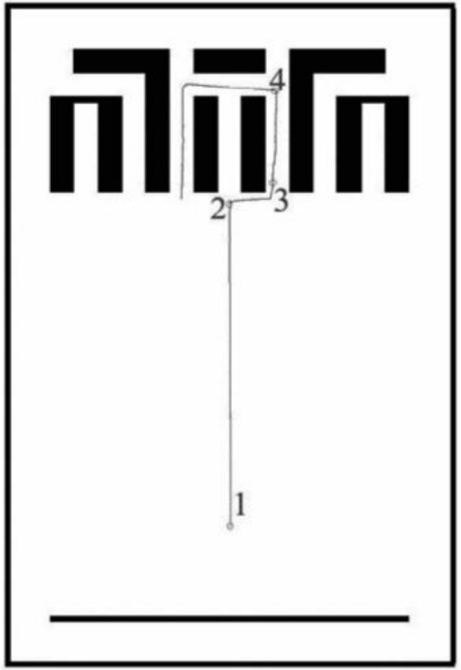
- SEE: The robot queries its sensors → finds itself next to a pillar
- ACT: Robot moves one meter forward
  - motion estimated by wheel encoders
  - accumulation of uncertainty
- SEE: The robot queries its sensors again → finds itself next to a pillar
- Belief update (information fusion)





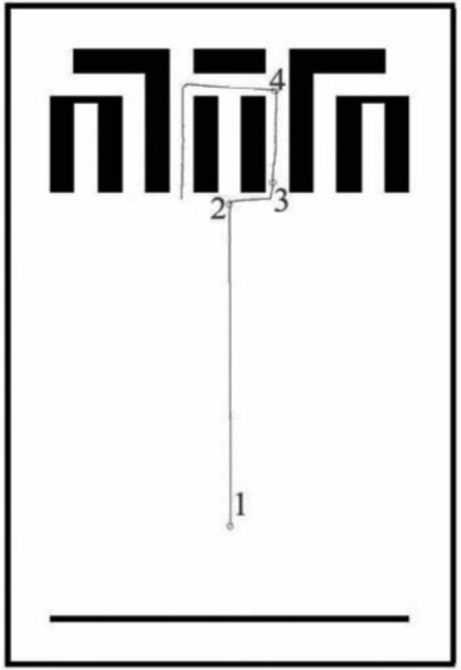
Error Propagation in Odometry
0.4
0.2
0
-0.2
-0.4
y [m]
0
0.5
1
x [m]



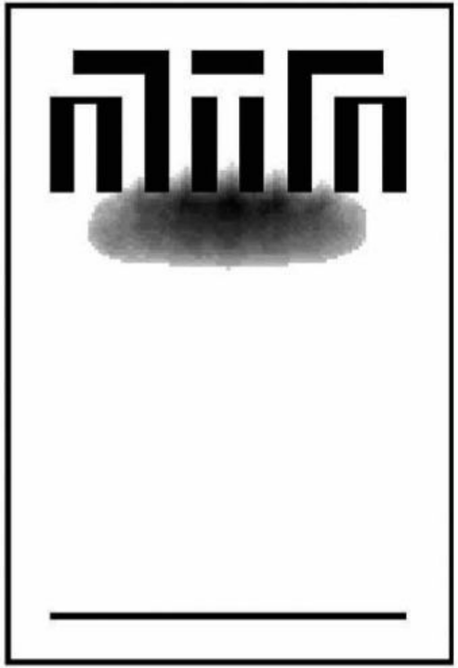


Path of the robot

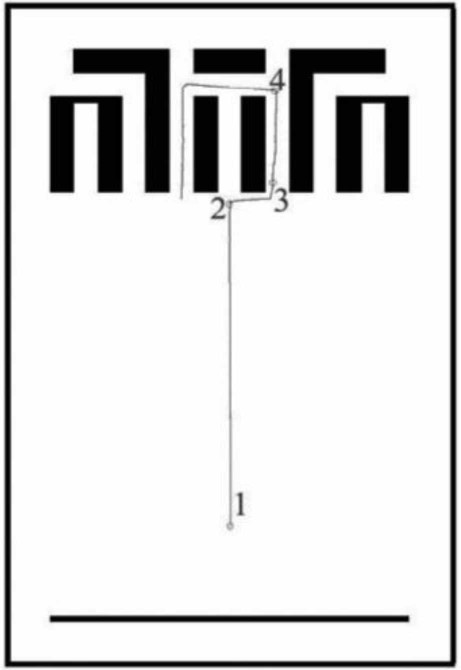
Belief states at positions 2, 3, and 4



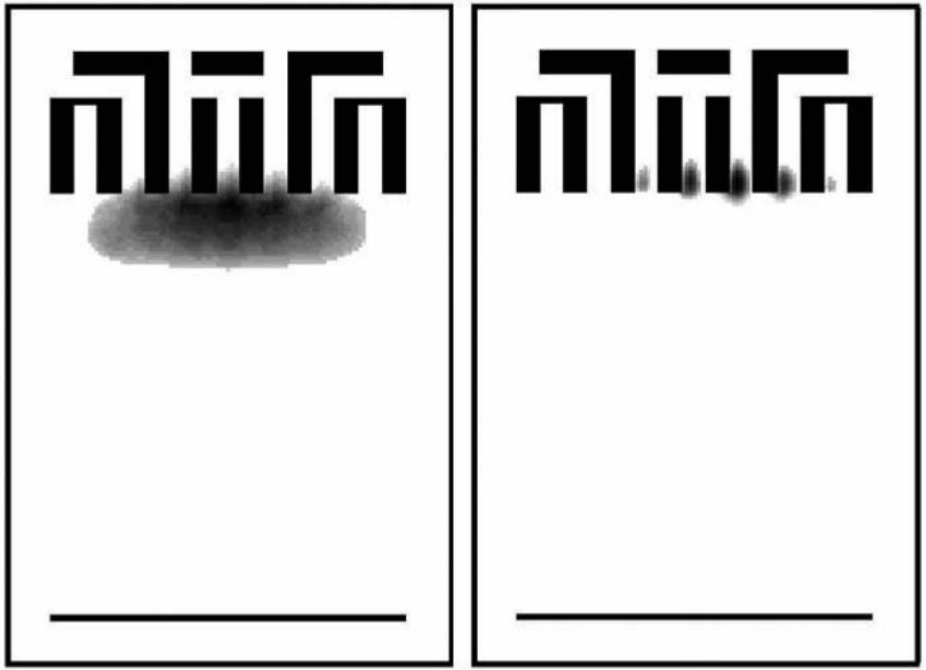
Path of the robot



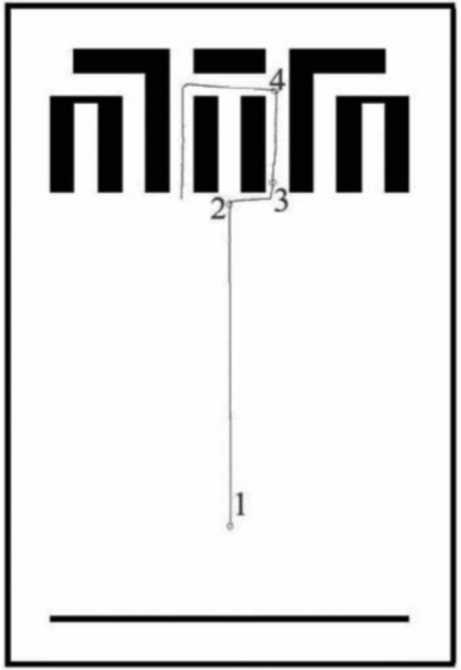
Belief states at positions 2, 3, and 4



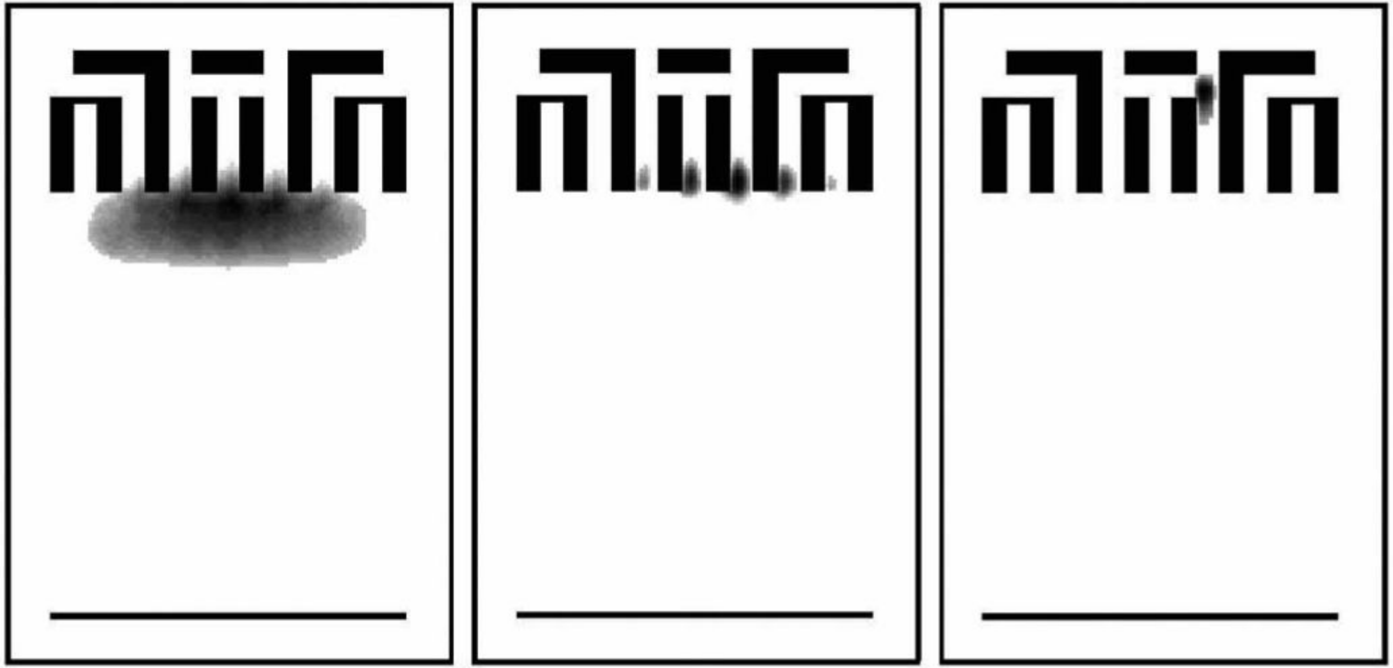
Path of the robot



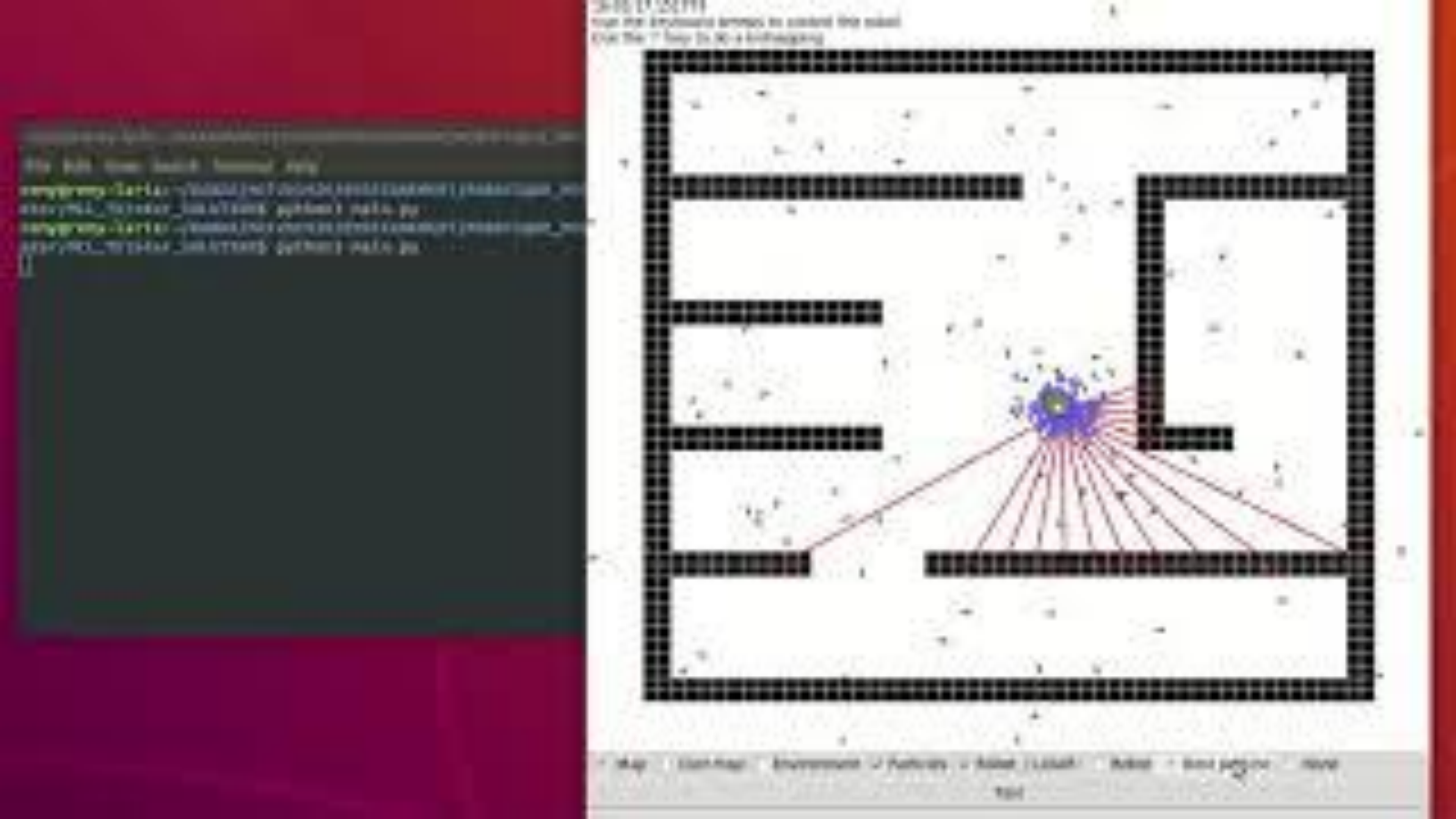
Belief states at positions 2, 3, and 4



Path of the robot



Belief states at positions 2, 3, and 4



Localization in simulation

Download archive Worlds.zip (updated version)

Uncompress it in your IR2121 repository folder

Launch a Gazebo simulation of the TurtleBot 3

Insert the model of the TI building in the simulation

Localization in simulation

```
sudo apt install
ros-humble-turtlebot3-navigation2
```

Launch AMCL with:

```
cd Worlds/scripts/ && ./launch_amcl.bash
```

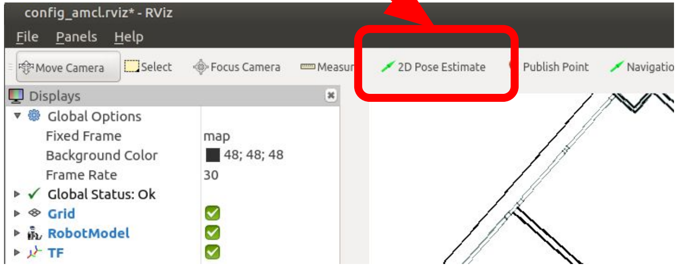
Launch RViz2 with:

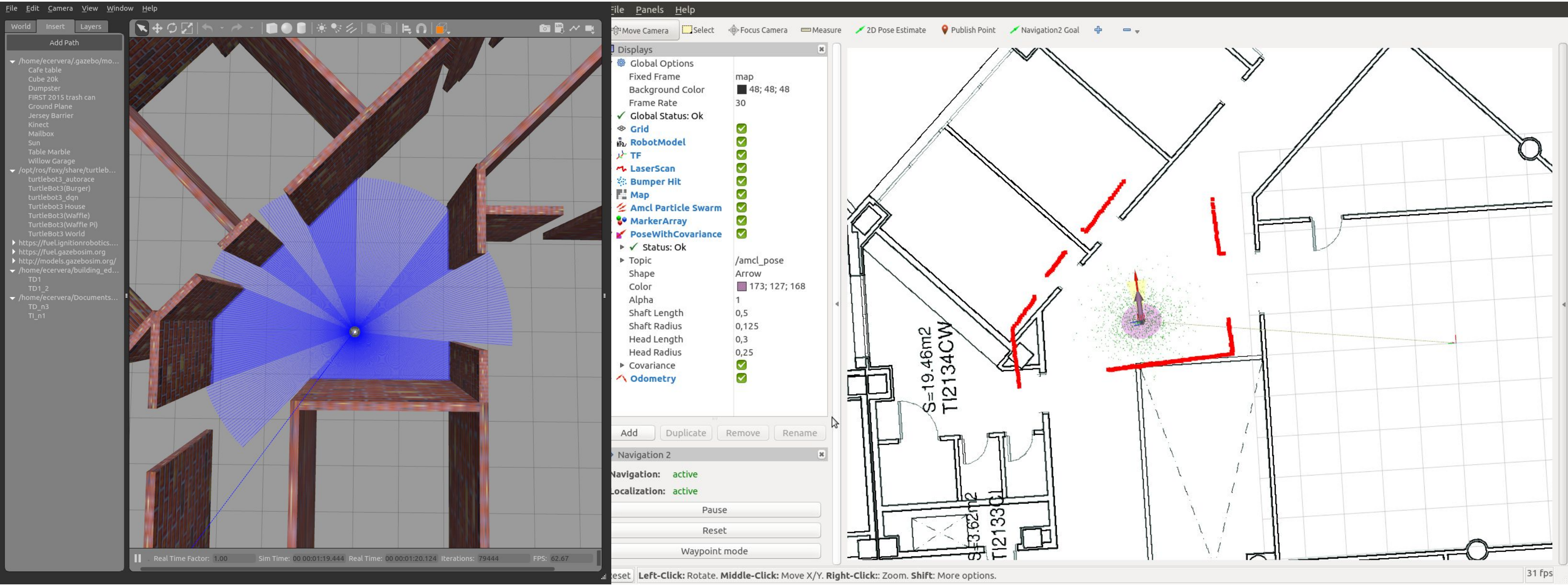
```
cd Worlds/scripts/ && ./visualize_localization.bash
```

```
#!/bin/bash
source /opt/ros/humble/setup.bash
export ROS_LOCALHOST_ONLY=1
export TURTLEBOT3_MODEL=burger

ros2 launch amcl.launch.py \
  use_sim_time:=True \
  map:=../TI_n1_edited.yaml
```

Estimate the 2D pose of the robot




File Edit Camera View Window Help
World Insert Layers
Add Path
/home/ecervera/.gazebo/mo...
Cafe table
Cube 20k
Dumpster
FIRST 2015 trash can
Ground Plane
Jersey Barrier
Kinect
Mailbox
Sun
Table Marble
Willow Garage
/opt/ros/foxy/share/turtleb...
turtlebot3_autorace
TurtleBot3(Burger)
turtlebot3_dqn
Turtlebot3 House
TurtleBot3(Waffle)
TurtleBot3(Waffle Pi)
TurtleBot3 World
https://fuel.ignitionrobotics...
https://fuel.gazebosim.org
http://models.gazebosim.org/
/home/ecervera/building_ed...
TD1
TD1_2
/home/ecervera/Documents...
TD_n3
TI_n1
Real Time Factor: 1.00 Sim Time: 00 00:01:19.444 Real Time: 00 00:01:20.124 Iterations: 79444 FPS: 62.67
File Panels Help
Move Camera Select Focus Camera Measure 2D Pose Estimate Publish Point Navigation2 Goal
Displays
Global Options
Fixed Frame map
Background Color 48; 48; 48
Frame Rate 30
Global Status: Ok
Grid
RobotModel
TF
LaserScan
Bumper Hit
Map
Amcl Particle Swarm
MarkerArray
PoseWithCovariance
Status: Ok
Topic /amcl_pose
Shape Arrow
Color 173; 127; 168
Alpha 1
Shaft Length 0,5
Shaft Radius 0,125
Head Length 0,3
Head Radius 0,25
Covariance
Odometry
Add Duplicate Remove Rename
Navigation 2
Navigation: active
Localization: active
Pause
Reset
Waypoint mode
S=19.46m2
TI2134CW
S=3.62m2
TI2133CL
Reset Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click:: Zoom. Shift: More options.
31 fps

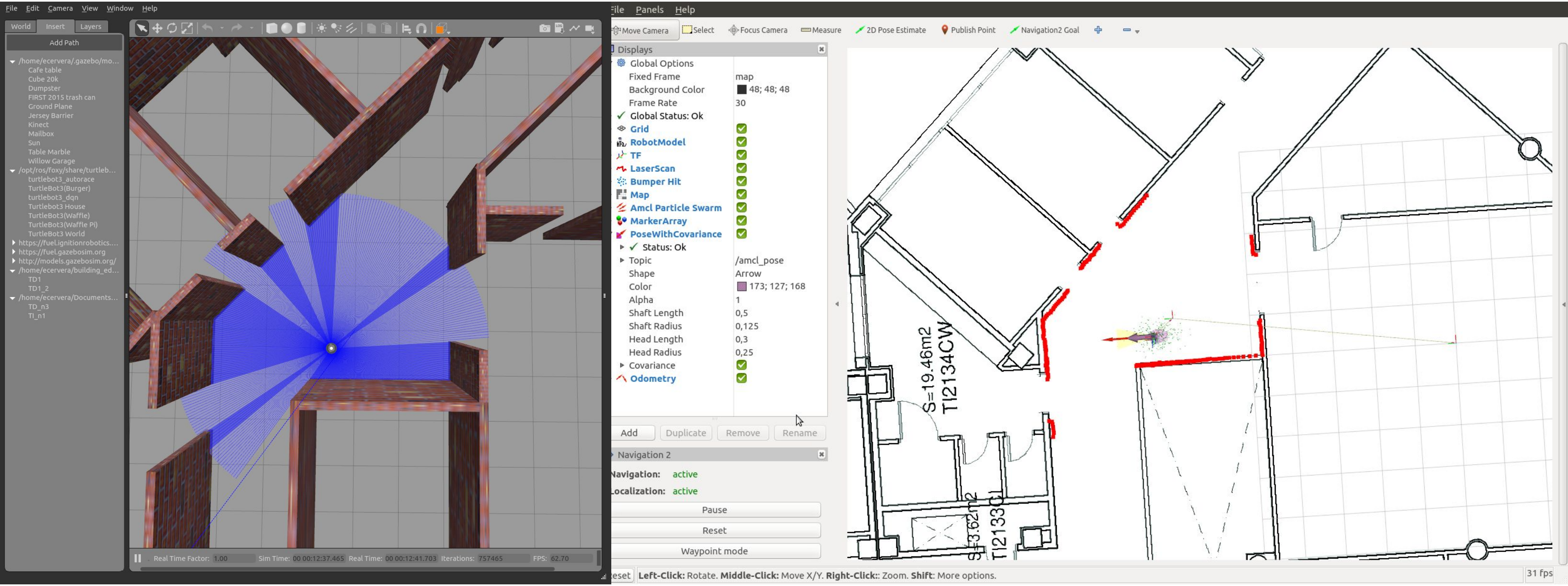
Localization in simulation

Move the robot around with the keyboard

The AMCL algorithm should automatically adjust the localization of the robot with respect to the map

Record the following topics in a bagfile for 60 seconds:

```
/clock /map /odom /scan /tf /tf_static
/amcl_pose /particle_cloud /robot_description
```


File Edit Camera View Window Help
World Insert Layers
Add Path
/home/ecervera/.gazebo/mo...
Cafe table
Cube 20k
Dumpster
FIRST 2015 trash can
Ground Plane
Jersey Barrier
Kinect
Mailbox
Sun
Table Marble
Willow Garage
/opt/ros/foxy/share/turtleb...
turtlebot3_autorace
TurtleBot3(Burger)
turtlebot3_dqn
Turtlebot3 House
TurtleBot3(Waffle)
TurtleBot3(Waffle Pi)
TurtleBot3 World
https://fuel.ignitionrobotics....
https://fuel.gazebosim.org
http://models.gazebosim.org/
/home/ecervera/building_ed...
TD1
TD1_2
/home/ecervera/Documents...
TD_n3
TI_n1
Real Time Factor: 1.00 Sim Time: 00 00:12:37.465 Real Time: 00 00:12:41.703 Iterations: 757465 FPS: 62.70
File Panels Help
Move Camera Select Focus Camera Measure 2D Pose Estimate Publish Point Navigation2 Goal
Displays
Global Options
Fixed Frame map
Background Color 48; 48; 48
Frame Rate 30
Global Status: Ok
Grid
RobotModel
TF
LaserScan
Bumper Hit
Map
Amcl Particle Swarm
MarkerArray
PoseWithCovariance
Status: Ok
Topic /amcl_pose
Shape Arrow
Color 173; 127; 168
Alpha 1
Shaft Length 0,5
Shaft Radius 0,125
Head Length 0,3
Head Radius 0,25
Covariance
Odometry
Add Duplicate Remove Rename
Navigation 2
Navigation: active
Localization: active
Pause
Reset
Waypoint mode
S=19.46m2
TI2134CW
S=3.62m2
TI2133C
Reset Left-Click: Rotate. Middle-Click: Move X/Y. Right-Click:: Zoom. Shift: More options.
31 fps

Verification of the recorded bagfile

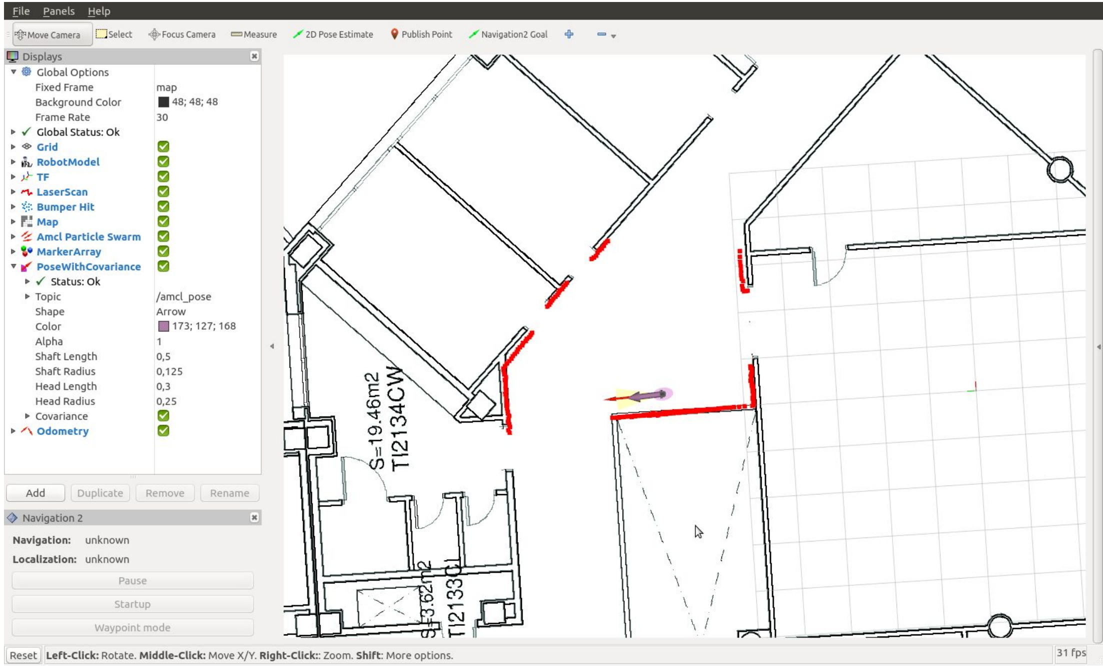
Stop the simulator

Stop RViz

Stop teleoperation

Launch RViz (./visualize_localization.bash)

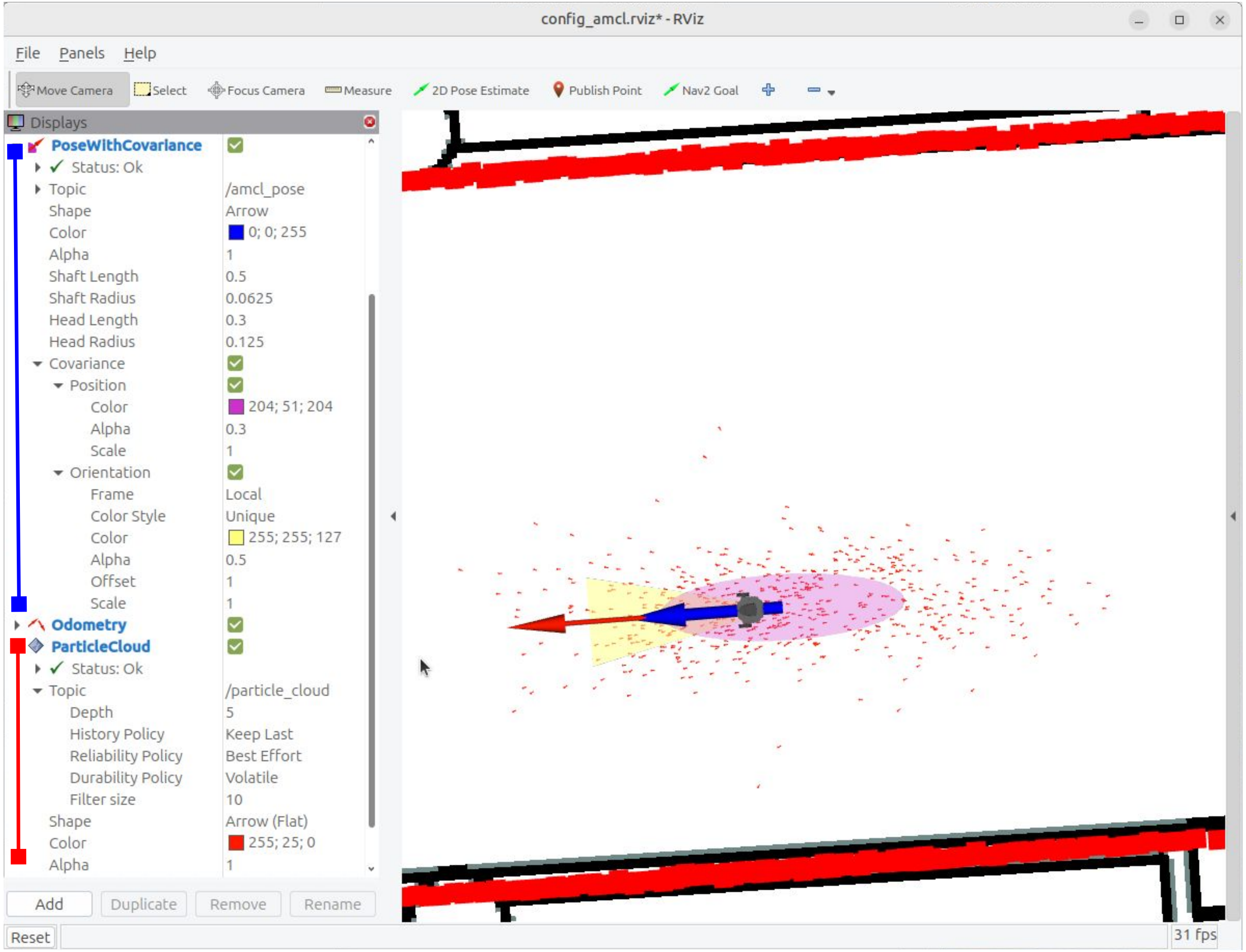
Play the bagfile



Visualization of AMCL pose and particle cloud

AMCL pose (with covariance)

Particle Cloud



Task 1: localization in simulation

Launch a Gazebo simulation of the TurtleBot 3

Insert your model of the TD building in the simulation

Launch AMCL

Launch RViz2

Estimate the 2D pose of the robot

Move the robot around with the keyboard

Record 60-90 seconds of the following topics in a bagfile:

```
/clock /map /odom /scan /tf /tf_static
/amcl_pose /particle_cloud /robot_description
```

Verification

Stop everything and relaunch rviz

Visualize the recorded bagfile prior to submission to Aula Virtual

If some topics are not displayed:

> Launch `ros2 bag record` **before** any other ROS script

Task 2: localization with the TurtleBot 3

Launch the TurtleBot 3 physical robot

Launch AMCL in the desktop PC

Launch RViz2 in the desktop PC

Estimate the 2D pose of the robot

Move the robot around with the keyboard

Record between 1 and 3 minutes of the following topics in a bagfile:

```
/map /odom /scan /tf /tf_static
/amcl_pose /particle_cloud /robot_description
```

Verification

Stop everything and relaunch rviz

Visualize the recorded bagfile prior to submission to Aula Virtual

If some topics are not displayed:

Launch `ros2 bag record` **before** any other ROS script

References

[Siegwart et al., 2011] Siegwart, R., Nourbakhsh, I. R., & Scaramuzza, D. (2011). Introduction to Autonomous Mobile Robots. MIT press. Catàleg UJI.

AMCL - ROS Wiki

AMCL — Nav2 1.0.0 documentation

Monte Carlo localization - Wikipedia

Robust Monte Carlo localization for mobile robots - ScienceDirect